

# 51.504 Machine Learning (2021)

## Homework 3

Due 9 Dec 2021, Thursday, 11.59pm  
Please submit on time for solutions to be released.

### Question 1

- (a) [10 points] What is the output when we convolve the left image with the  $3 \times 3$  filter on the right (assuming stride = 1 and no padding)?

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 1 | 2 | 0 | 3 |
| 2 | 4 | 2 | 1 | 1 |
| 1 | 1 | 5 | 3 | 0 |
| 1 | 2 | 6 | 0 | 1 |
| 0 | 1 | 2 | 0 | 3 |

 $\times$ 

|    |   |    |
|----|---|----|
| 1  | 0 | -1 |
| 0  | 1 | 0  |
| -1 | 0 | 1  |

 $=$ 

|  |  |  |
|--|--|--|
|  |  |  |
|  |  |  |
|  |  |  |

- (b) [10 points] What is the output if we instead perform convolution using padding = 1, stride = 2, and then apply ReLU?

### Question 2

Let  $X = (X_1, X_2, X_3, X_4)$  be the random variable with  $0 \leq X_i \leq 4$  for  $i = 1, 2, 3, 4$ , and density function

$$f_X(x) = \frac{1}{Z_1} \exp\{x_1 x_2 x_3 x_4\}, \quad Z_1 = \int_0^4 \int_0^4 \int_0^4 \int_0^4 \exp\{x_1 x_2 x_3 x_4\}.$$

Let  $Y = (Y_1, Y_2, Y_3, Y_4)$  be the random variable with  $0 \leq Y_i \leq 4$  for  $i = 1, 2, 3, 4$ , and density function

$$f_Y(y) = \frac{1}{Z_2} \exp\{y_1 y_2 y_3 + y_1 y_2 + y_4\}, \quad Z_2 = \int_0^4 \int_0^4 \int_0^4 \int_0^4 \exp\{y_1 y_2 y_3 + y_1 y_2 + y_4\}.$$

Let  $G$  be the graph in Figure 1.

- (a) [5 points] Show that  $G$  is a Markov random field for  $Y$  but not  $X$ .

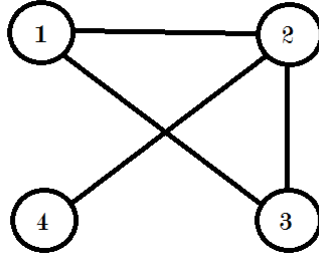


Figure 1: Graph  $G$

- (b) [5 points] Write down a conditional independence relationship for  $Y$  that can be read off the graph  $G$  due to the global Markov property. (Note that  $Y_i$  is the random variable associated with node  $i$ .)
- (c) [5 points] Write down the conditional independence relationship for  $Y$  that follows from the local Markov property for node 4.
- (d) [5 points] Draw the appropriate arrows on  $G$  for the Bayesian network associated with the following set of structural equations:

$$\begin{aligned}
 Y_1 &= -2Y_3 + \epsilon_1, \\
 Y_2 &= 4Y_1 + Y_3 + \epsilon_2, \\
 Y_3 &= \epsilon_3, \\
 Y_4 &= -3Y_2 + \epsilon_4, \\
 \epsilon_i &\sim \mathcal{N}(0, 1).
 \end{aligned}$$

### Question 3

Consider a mixture of two Gaussians,  $\mathcal{N}(\mu_1, \sigma_1^2)$  with density function  $f_1(x)$  and proportion  $\pi_1$ , and  $\mathcal{N}(\mu_2, \sigma_2^2)$  with density function  $f_2(x)$  and proportion  $\pi_2$ . The EM algorithm iteratively updates the parameters with the E-step,

$$\gamma(z_k^{(i)}) = \frac{\pi_k C_k \exp\{-(x^{(i)} - \mu_k)^2 / 2\sigma_k^2\}}{\sum_{j=1}^2 \pi_j C_j \exp\{-(x^{(i)} - \mu_j)^2 / 2\sigma_j^2\}},$$

and the M-step,

$$\begin{aligned}\pi_k &= \frac{1}{2} \sum_{i=1}^2 \gamma(z_k^{(i)}), \\ \mu_k &= \frac{\sum_{i=1}^2 x^{(i)} \gamma(z_k^{(i)})}{\sum_{i=1}^2 \gamma(z_k^{(i)})}, \\ \sigma_k^2 &= \frac{\sum_{i=1}^2 (x^{(i)} - \mu_k)^2 \gamma(z_k^{(i)})}{\sum_{i=1}^2 \gamma(z_k^{(i)})}.\end{aligned}$$

Suppose we perform EM on the sample data set  $\{1, 1.5\}$ . Let our initial parameter estimates be  $\mu_1 = 0, \mu_2 = 3, \sigma_1 = 1, \sigma_2 = 2$ . Also let our initial mixture parameters be  $\pi_1 = 1/3$ . We then have  $f_1(1) = 0.241, f_1(1.5) = 0.13, f_2(1) = 0.12$  and  $f_2(1.5) = 0.15$ .

- (a) [10 points] Calculate the probabilities that each data point belongs to each Gaussian given the initial parameters.
- (b) [10 points] Compute the next iteration of  $\pi_1, \pi_2, \mu_1, \mu_2$ .
- (c) [10 points] Compute the next iteration of  $\sigma_1^2$  and  $\sigma_2^2$

## Question 4

The exponential recency-weighted average update is

$$Q_n = Q_{n-1} + \alpha (r(n) - Q_{n-1}), \quad \alpha \in (0, 1],$$

and we can expand the expression to get

$$Q_n = (1 - \alpha)^n r(0) + \alpha(1 - \alpha)^{n-1} r(1) + \cdots + \alpha(1 - \alpha) r(n-1) + \alpha r(n). \quad (1)$$

- (a) [10 points] If we write the above expression as

$$Q_n = \beta_0 r(0) + \beta_1 r(1) + \cdots + \beta_{n-1} r(n-1) + \beta_n r(n),$$

use the geometric series formula  $\sum_{k=0}^{n-1} \alpha r^k = \alpha \left( \frac{1-r^n}{1-r} \right)$  to show that

$$\sum_{k=0}^n \beta_k = 1.$$

- (b) [10 points] Let  $w = (w(0), w(1), \dots, w(T))$  be a vector of weights. We assume  $w(0) = 1$ , and let  $S(n) := \sum_{i=0}^n w(i)$  for  $n = 0, 1, \dots, T$ . We assume  $S(n)$ , for all  $n$ , has been pre-computed and stored in an array, so it only requires constant time to access their values.

Now, we define a new update rule as follows:

$$Q_n = \frac{\sum_{i=0}^n w(i)r(i)}{S(n)}. \quad (2)$$

Instead of having to compute the sum in the numerator at each step, give and prove an incremental update formula to efficiently compute  $Q_n$  using just  $Q_{n-1}$ ,  $w(n)$ ,  $r(n)$ ,  $S(n)$  and  $S(n-1)$ .

- (c) [10 points] It is possible to set  $w(i)$  such that (2) is exponentially weighting the most recent rewards by a factor of  $\alpha$ , provided that  $\alpha < 1$ . For  $i \geq 1$ , give and prove a formula for  $w(i)$  using just  $\alpha$  and  $i$  such that (2) is equivalent to the exponential recency-weighted update rule.

**Hint:** Compare the coefficients of  $r(i)$  in (1) and (2), and use the fact that  $w(0) = 1$  and part (a) to compute  $S(n)$ .